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### Reflective Journal – AI 900 Lab 4: Azure AI Document Intelligence

In this lab, I explored Azure AI Document Intelligence within the Azure AI Foundry portal to analyze and extract data from a receipt using prebuilt AI models. This process demonstrated how AI can enhance traditional Optical Character Recognition (OCR) by structuring extracted data for easier processing and storage.

#### Key Takeaways:

- AI-powered OCR goes beyond simple text extraction. Unlike basic OCR, Document Intelligence identifies key data fields such as merchant names, addresses, and transaction details, making the extracted information more structured and useful.
- Prebuilt models in Azure AI Foundry automate document processing. Instead of manually entering receipt details, AI can quickly categorize and extract relevant information.

- Confidence scores provide a way to assess accuracy. Each extracted field is assigned a probability, helping to determine whether data should be manually reviewed before integration.

I approached this lab differently by first trying to navigate Azure AI Foundry on my own before following the step-by-step instructions. This was a way to test my intuition and ability to explore a new platform. While Azure still doesn't feel completely familiar, I noticed that my ability to navigate unfamiliar tools is improving. I want to develop the skill of quickly finding my way around new platforms, as that's essential in technology-driven environments.

One part of this lab that stood out was how relevant AI-powered document processing is to real-world applications. It reminded me of my past experience with Concur, an expense reporting platform I used long ago. Back then, the process was very manual, and I avoided it until absolutely necessary. I hope they've integrated AI-powered OCR by now—if not, someone should and sell it to them. Automating receipt processing would make expense reports significantly easier and more efficient.

#### Final Thoughts:

- AI-powered document intelligence has clear applications in finance, retail, and other industries where document processing is critical.
- Automating manual data entry reduces errors and improves efficiency.

- Learning to navigate new platforms is a valuable skill, and I want to continue improving my ability to adapt quickly to different tools.

Overall, this lab reinforced the importance of structured data extraction for automation and highlighted how AI can improve everyday business processes. AI-powered tools like Document Intelligence can save time, reduce errors, and enhance productivity, making them valuable for businesses that deal with large volumes of unstructured data.

